

Pénélope Azuelos

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Employment

2026 – present Queens' College, University of Cambridge, UK
Junior Research Fellow

Education

2022 – 2026 University of Bristol, UK
PhD in Mathematics, supervised by Mark Hagen and Indira Chatterji

2021 – 2022 École Normale Supérieure de Lyon, France
Master (M2) Mathématiques Avancées

2017 – 2021 University of Bristol, UK
Master in Science in Mathematics (with first class honours)

Publications

A guide to constructing free transitive actions on median spaces, Advances in Mathematics, 503, 2026.

(with Damien Gaboriau) *Perfect kernel and dynamics: from Bass-Serre theory to hyperbolic groups*, Mathematische Annalen, 391: 4733–4789, 2025.

On subgroups with narrow Schreier graphs, Bulletin of the London Mathematical Society, 56(12): 3652–3668, 2024.

Preprints

(with Mark Hagen) *Periodic quasiflats in hierarchically hyperbolic spaces*, arXiv:2608.01513.

Uncountably many homogeneous real trees with the same valence, arXiv:2511.03722.

Teaching

Teaching assistant at the University of Bristol for the following courses:

Spring 2024 Introduction to proofs and group theory MATH10010

Autumn 2023 Combinatorics MATH30030

Spring 2020 Introduction to geometry MATH20004

Conference Talks

August 2026 Groups, Dynamics and Operator Algebras Conference, University of Glasgow, UK

June 2026 Groups acting on non-positively curved spaces, International Academic Forum, Heidelberg, Germany

- Branch groups: subgroups, rigidity, topologies, Universidad Complutense de Madrid, Spain
- May 2026 Geometric group theory around cube complexes, CIRM, France
- April 2026 Coarse geometry of groups, Centre Paul-Langevin, Aussois, France
- February 2026 Median Geometry and Applications, MFO, Oberwolfach, Germany.
- Orbit meetings, University of Warwick, UK
- November 2025 Parole aux jeunes scientifiques du RT tétraèdre de Platon, Montpellier, France.
- Operator algebras in the south of England, University of Bristol, UK.
- June 2025 Conference on hyperbolicity and generalizations, Harbin Institute of Technology, China.
- January 2025 RSME's 7th congress of young researchers, parallel session on geometric group theory, UPV/EHU, Bilbao, Spain.
- September 2024 World of Geometric Group Theory, online.
- March 2024 Rigidité en théorie des groupes : aspects géométriques et ergodiques, Université Paris-Saclay, Orsay, France.
- May 2023 Measured Group Theory, Stochastic Processes on Groups and Borel Combinatorics, CIRM, Marseille, France.

Seminar Talks

- May 2026 Geometric group theory seminar, University of Oxford, UK
- April 2026 Junior geometry seminar, University of Bristol, UK
- March 2026 Junior topology and group theory seminar, University of Oxford, UK
- February 2026 DAGGER seminar, University of Warwick, UK.
- Junior Geometry Colloquium, KIT, Karlsruhe, Germany.
- January 2026 Geometry and Topology Seminar, University of Bristol, UK.
- October 2025 Junior Algebra and Combinatorics Seminar, University of Bristol, UK.
- September 2025 Séminaire Groupes et Actions, Université Paris-Saclay, Orsay, France.
- February 2025 Junior Geometry Seminar, University of Bristol, UK.
- December 2024 Séminaire d'Algèbres d'Opérateurs, IMJ-PRG, Paris, France.
- October 2024 Séminaire de l'équipe d'algèbre, topologie et géométrie, Laboratoire J.A. Dieudonné, Nice, France.
- September 2024 Séminaire de Théorie Ergodique, Institut de recherche mathématique de Rennes, France.
- Algebra Geometry and Topology seminar, Heriot-Watt University, Edinburgh, UK.
- May 2024 International Young Seminar on Bounded Cohomology and Simplicial Volume, online.
- March 2024 GdT Action ! ENS de Lyon, France.
- February 2024 Junior Geometry Seminar, University of Cambridge, UK.

December 2023	Postgraduate Seminar, University of Bristol, UK.
November 2023	Junior Topology and Group Theory Seminar, University of Oxford, UK.
September 2023	Le séminaire virtuel francophone Groupes et Géométrie, online.
January 2023	Junior Geometry Seminar, University of Bristol, UK.

Service

2025	Organisation of the Postgraduate Group Theory Conference 2025
2023 – 2024	Organisation of the Bristol Junior Geometry Seminar

Outreach

July 2026	Talk and panel participation at an undergraduate summer school for women in mathematics, University of Warwick
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Awards and funding

2024	Grant from the Académie d'Excellence "Systèmes Complexes" for a three month research visit to the LJAD in Nice, supervised by Indira Chatterji.
2022 – 2026	University of Bristol PhD scholarship.
2021 – 2022	MiLyon scholarship for an M2 masters program.
2020	Summer research bursary funded by the Heilbronn Institute for Mathematical Research, supervised by Misha Rudnev.

Languages

English and French (native speaker)